Chapter 8
Incremental Cash Flow

For Multiple Alternatives



Why is it more complicated?

Sarah's father gives her a gift of $10,000 and motivates her to put it in a savings account with a guaranteed APR of 3%. However, Sarah, having taken the Engineering Economy course, learned that she can do better, but with some risks. Hence, she is evaluating two investment alternatives:

A. Peer-to-peer lending opportunity that requires an investment of $5,000 and has an expected $i^* = 20\%$ per year.
B. Investment in a stock portfolio that requires $9,000 and has an expected $i^* = 15\%$



What is an Incremental Cash Flow

- The difference of two cash flows:

Incremental Cash Flow = Cash Flow with higher investment – Cash Flow with lower investment

Alternatives require the equal-service assumption

- i^* of each alternative > MARR

Only for **Revenue** Alternatives

+

- If i^* of the *incremental cash flow* > MARR → Select alternative with higher investment

Why?



Incremental Cash Flow Method

- Mutually exclusive alternatives 
- Independent Projects 
- Cost alternatives
  - Disregard the i^* of each alternative 

Incremental Cash Flow - Example

- A company is evaluating two mutually exclusive machines.
- The MARR is 10% per year. The cash flow details are as follows:

	Machine A	Machine B
First Cost	-$10,000	-$25,000
Annual Revenue	$5,000	$9,000
Salvage Value	$2,000	$5,000
Life (years)	5	5

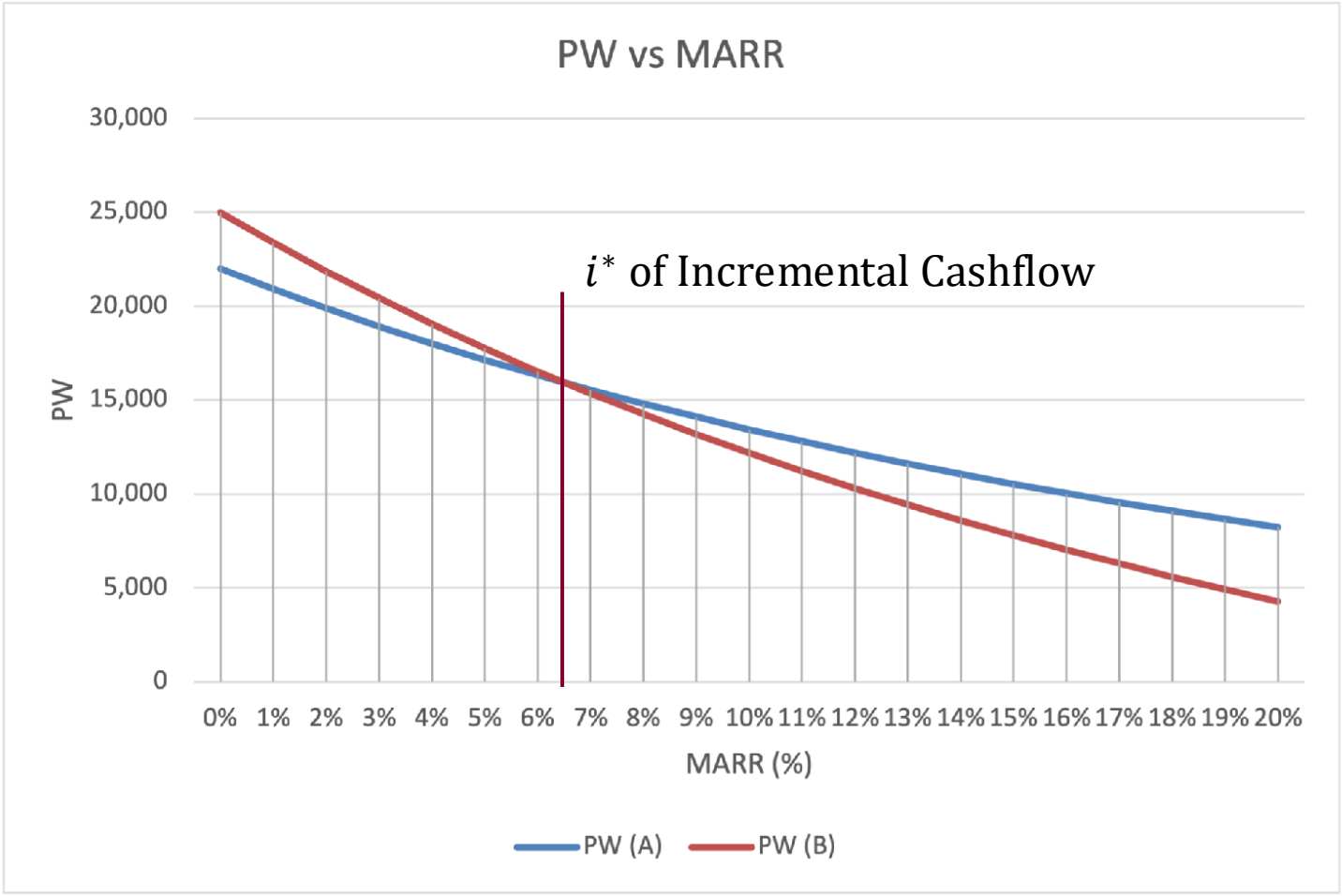
Microsoft Excel Worksheet

Different Lifetimes

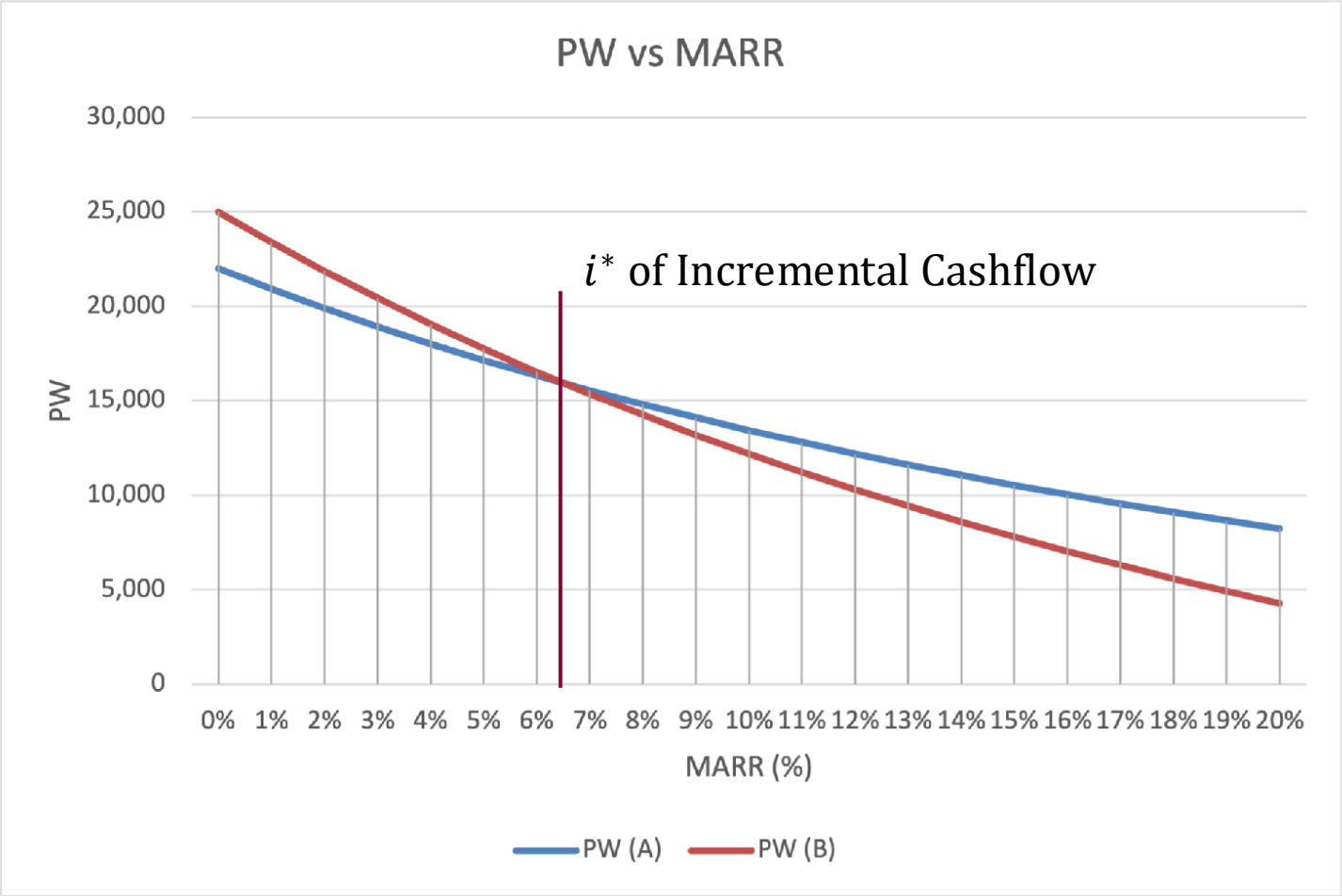
	Machine A	Machine B
First Cost	-$8,000	-$20,000
Annual Revenue	$6,000	$7,000
Salvage Value	$1,000	$3,000
Life (years)	3	6

LCM is needed → Incremental Cash Flow

Understanding what is happening!!!



Understanding what is happening!!!



- The curves intersect at exactly MARR = 6.54%, which is the incremental rate of return
- For any MARR < 6.54%, the PW of Machine B is higher than Machine A
- For any MARR>6.54% (including the company's actual MARR of 10%), the PW of Machine A is higher.

Incremental ROR – Multiple Alternatives

1. Rank alternatives in **ascending** order of initial investment.
2. Establish the "Do Nothing" (DN) alternative as the baseline (if applicable) OR the lowest-cost alternative as the first defender.
3. Compare the defender against the next alternative in the rank (the challenger) using **incremental cash flow** (ΔPW).
    1. If $\Delta i^* \geq MARR$, the challenger wins and becomes the new defender. Otherwise, the current defender remains.
4. Repeat until all alternatives are evaluated.

Higher – Lower Init. Inves.



Example – MARR=10%

Alternative	Initial Cost (1000 USD)	Annual Revenue (1000 USD)	Life	ROR (%)
A	-10	6	2	13.07
B	-15	5.5	3	4.92
C	-20	9	3	16.65
D	-30	8	6	15.34

1. Discard alternative with ROR < MARR
2. Rank alternatives in **ascending**
3. ~~Establish the "Do Nothing" (DN) alternative as the baseline (if applicable) OR the lowest-cost alternative as the first defender.~~ Not needed given individual ROR
4. Compare the defender against the next alternative in the rank (the challenger) using **incremental cash flow** (ΔPW).
    1. If $\Delta i^* \geq MARR$, the challenger wins and becomes the new defender. Otherwise, the current defender remains.
5. Repeat until all alternatives are evaluated.

Guess who's Back!!

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AW analysis for Incremental Cashflows

